Declaration of AI assistance

European Journal of Combinatorics

Author-prepared disclosure; not an official publisher form or signature.

Manuscript. Two conjectures of Wormald on Hamilton cycles in tournaments

Human authors. Deep Bhattacharjee.

Claude through Claude Code assisted the original literature search, calculations, verification programs, Lean development, figure-generation code and text. OpenAI Codex assisted this revision's proof and literature audit, explicit analytic derivatives, quantitative nondegeneracy corollary, direct spectral-radius proof, exact matrix checker and submission files. The author retains responsibility for the mathematics, references and final submission decisions.

Human oversight. The submitting author stated on 25 October 2026 that the text is a mix of human-written and AI-generated material and that all parts have been checked and understood by a human. This records that author declaration. Human authors retain responsibility for mathematical accuracy, citations and the final version; AI systems are not human coauthors.

Verification scope. The unchanged Lean proof tree has a complete successful build and 23 axiom reports using only propext, Classical.choice and Quot.sound. It formalizes the determinant algebra and cyclic limit after identifying H with det(E^T F E). The analytic identification, asymptotic expansion, spectral bounds and new quantitative corollary have written proofs and are not claimed as Lean-formalized. Fresh exact checks verify 41 full Hessian matrices (3136 entries) and their bounds, and 142 manuscript values; numerical checks complement the universal proofs.

Supporting record. The separate AI_PROVENANCE.md identifies the documented tasks and tools. No unprovided model version, human signature, confidential transcript or journal approval is asserted.

Policy reference. https://www.elsevier.com/about/policies-and-standards/generative-ai-policies-for-journals

Prepared on 25 October 2026. The author reviews this statement and completes any additional live-portal declarations before submission.